

Yi Cui

PhD Student in AI · Trustworthy Verification of Multimodal Reasoning & Agentic Systems

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EDUCATION

Ph.D. in Artificial Intelligence

2024 – Present

The Hong Kong University of Science and Technology (Guangzhou)

Supervisor: Prof. Hui Xiong — Founding Head of AI Thrust (HKUST-GZ); Fellow of ACM, IEEE, AAAS, AAAI, CAAI

M.S. in Computer Engineering (GPA 3.5/4.0)

2024

New York University

B.S. in Computer Science (GPA 3.4/4.0; Minor in Economics)

2021

Pennsylvania State University

RESEARCH

My research builds toward trustworthy, process-level verification of multimodal and agentic systems. Starting from benchmark construction that exposes how i.i.d. evaluation overstates real-deployment reliability (SafeBuild-Bench, KDD'26), I moved to the verification layer itself — diagnosing a hidden position-prior confounder that inflates multimodal process-reward scores — and then to GLANCE, which separates answers that genuinely use visual evidence from ones that are correct by prior, without training, labels, or edited images. I am now extending accountability evaluation from perception to long-horizon agent behavior (RedDust), and mapping 100+ papers from visual chain-of-thought to verifiable reasoning agents into a process-level verification agenda (survey in preparation).

PUBLICATIONS

† Co-first author * Corresponding author

- [1] **SafeBuild-Bench: A Temporal-Robust Construction Safety Benchmark with Graph-Enhanced Data Mining.** Yi Cui†, Zilin Wang†, Yijie Xu, Qianyi Cai, Huizai Yao, Shuai Jiang, Bingzhuo Zhong*, Hui Xiong*. **ACM SIGKDD 2026 (KDD)**, Datasets & Benchmarks Track — Accepted.
- [2] **Beyond Boundaries: Leveraging Vision Foundation Models for Source-Free Object Detection.** Huizai Yao, Sicheng Zhao, Pengteng Li, Yi Cui, Shuo Lu, Weiyu Guo, Yunfan Lu, Yijie Xu, Hui Xiong*. **AAAI 2026** — Accepted.
- [3] **Contrastive Graph Matching for Domain Adaptive Object Detection.** Pengteng Li, Huizai Yao, Yi Cui, Yunfan Lu, Yijie Xu, Minghao Yang, Pinhao Song. **arXiv** — Preprint.

MANUSCRIPTS UNDER REVIEW & IN PREPARATION

- [4] **Beyond Step AUC: Hidden Position Priors and Capability Trade-offs in Multimodal Process Reward Models.** Yi Cui, Huizai Yao, Zilin Wang, Zihan Wang, Hui Xiong*. Under review, **ACL 2027**.
Published multimodal-PRM rankings turn out to be driven by a hidden position prior rather than step-verification ability: on VisualProcessBench a step-index-only baseline reaches 0.73 AUC while the best trained PRM reaches 0.36. Proposes ability-specific metrics against a position-controlled baseline.
- [5] **Grounding or Guessing? A Privileged-Crop Detector of Visual Evidence Use in Multimodal LLMs (GLANCE).** Yi Cui, Hui Xiong*. In preparation, targeting **ICLR 2027**.
Estimates, from an untouched image, whether a multimodal LLM used the visual evidence or was merely right by prior — scoring one answer under a crop, the full image, and a masked view, with no training and no labels. Separates grounded from guessed-correct at 0.81 AUROC, +0.26 over the strongest region-blind baseline.
- [6] **RedDust: Short-Horizon Social Skill Does Not Predict Long-Horizon Consistency in LLM Agents.** Yi Cui, Hui Xiong*. In preparation, targeting **ICLR 2027**.
Independent project — a 30-day agent-accountability benchmark whose cross-model panel shows that short-horizon social skill does not predict long-horizon consistency: at matched early comprehension, model families split between keeping and breaking their own commitments.

PROJECTS

Construction-Site Safety-Hazard VLM (Guangdong Heavy Industry, Industry Collaboration) 2025 – Present
Qwen3-VL-32B student (LoRA) · Qwen3-VL-235B-A22B teacher (distillation) · Contrastive labeling · GEMS · LLaMA-Factory · 8×A6000 · Agentic RAG · vLLM

- Led the data pipeline, model adaptation, and evaluation end to end for a pilot VLM system that flags construction-site safety hazards, cites the violated regulation, and drafts rectification advice.
- Raised hazard-recognition accuracy by ~35% (relative) over baseline on the internal validation set across v2–v5 iterations — driven mainly by data-quality refinement (semantic extraction over expert annotations into a sharper hazard taxonomy) and system-prompt engineering (per-category definitions + a fixed output schema).
- Distilled a same-family Qwen3-VL-235B-A22B teacher into a Qwen3-VL-32B student via LoRA — the largest student our compute budget supports — using a contrastive teacher process to auto-generate structured “description → hazard → regulation → fix” labels and sharply cut manual labeling cost.
- Curated an expert-reviewed 30K-instance SFT dataset from 100K+ raw inspection records, using GEMS (graph-enhanced data selection) to counter severe class long-tail and focus senior-engineer review on high-value data instead of the full corpus; expanded the hazard taxonomy from 62 to 90 classes along the way.
- Designing a decoupled Agentic RAG pipeline (in progress) to replace direct VLM generation of regulation citations, targeting traceability and maintainability.

SKILLS

Research Areas: Vision-Language Models (VLM/MLLM) · Multimodal Evaluation & Benchmarking · Process Reward Models (PRM) / Verifiable Reasoning · Data-Centric AI

Training & Methods: LoRA / PEFT · Knowledge Distillation · Contrastive Learning · Distributed Training (multi-GPU) · Agentic RAG

Programming: Python · C++ · Java **Frameworks/Tools:** PyTorch · MATLAB · React/Phaser · TypeScript · vLLM · LLaMA-Factory

TEACHING

Teaching Assistant, HKUST (Guangzhou)

AIAA 3053 — Reinforcement Learning: Principles and Methods UFUG 2104 — Applied Statistics

LANGUAGES

English (Fluent — 6 years of study in the U.S.) Chinese (Mandarin, Native)